Copernicus Global Land Operations

"Vegetation and Energy"

"CGLOPS-1"

Framework Service Contract N° 948120 – IPR - 2023

PRODUCT USER MANUAL

SENTINEL-3 OLCI AND SLSTR SURFACE REFLECTANCE

VERSION 2.3

Issue I1.10

Organization name of lead contractor for this deliverable: VITO

Book Captain: Dominique De Munck (VITO)
Contributing Authors: Carolien Toté, Davy Wolfs (VITO)
Dominique Jolivet (HYGEOS)
Jorge Sanchez-Zapero (EOLAB)

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List of Acronyms

Acronym	Meaning
6S	Second Simulation of the Satellite Signal in the Solar Spectrum
AC	Atmospheric Correction
AD	Applicable Document
AOI	Area Of Interest
AOT	Aerosol Optical Thickness
ATBD	Algorithm Theoretical Basis Document
B	Mean bias
BRDF	Bi-directional Reflectance Distribution Function
CA	Contracting Authority
CAL/VAL	Calibration & Validation
CAMS	Copernicus Atmosphere Monitoring Service
CEOS	Committee of Earth Observation Satellites
CF	Climate & Forecast
CGLOPS	Copernicus Global Land Operations
CLMS	Copernicus Land Monitoring Service
COG	Cloud Optimized GeoTIFF
DMP	Dry Matter Productivity
DN	Digital Number
ECV	Essential Climate Variable
EO	Earth Observation
EOLAB	Earth Observation Laboratory
EPSG	European Petroleum Survey Group
ESA	European Space Agency
EU	European Union
FAPAR	Fraction of Absorbed Photosynthetically Active Radiation
FCOVER	Fraction of Vegetation Cover
FWC	Framework Contract
GBOV	Ground-Based Observations for Validation
GCOS	Global Climate Observing System
GDAL	Geospatial Data Abstraction Library
HYGEOS	Hydrogeology and Earth Observations
HYPERNETS	HYPERspectral radiometer integrated in automated NETworkS
IdePix	Identification of Pixels
JRC	Joint Research Centre
LAI	Leaf Area Index
LEO	Low Earth Orbit
LPV	Land Production Validation

MERRA-2	Modern-Era Retrospective analysis for Research and Applications, Version 2
NDVI	Normalized Difference Vegetation Index
NetCDF	Network Common Data Form
NPP	Net Primary Production
NRT	Near Real Time
OGC	Open Geospatial Consortium
OLCI	Ocean and Land Colour Instrument
OPT-MPC	Optical Mission Performance Cluster
PQAD	Product Quality Assurance Document
PUM	Product User Manual
QAR	Quality Assessment Report
RadCalNet	Radiometric Calibration Network
RAL	Rutherford Appleton Laboratory
RMSD	Root Mean Square Deviation
S3	Sentinel-3
SAA	Sun Azimuth Angle
SLSTR	Sea and Land Surface Temperature Radiometer
SMAC	Simplified Model for Atmospheric Corrections
SNAP	Sentinel Application Platform
SWIR	Short-Wave Infrared
SZA	Sun Zenith Angle
TN	Technical Note
TOA	Top Of the Atmosphere
TOC	Top Of Canopy
VAA	Viewing Azimuth Angle
VIS	Visible
VITO	Vlaamse Instelling Voor Technologisch Onderzoek
VR	Validation Report
VZA	Viewing Zenith Angle
WGS	World Geodetic System
WKT	Well-Known Text format

EXECUTIVE SUMMARY

The Copernicus Land Monitoring Service (CLMS) produces a series of qualified bio-geophysical products on the status and evolution of the land surface. The products are used to monitor vegetation, crops, water cycle, energy budget and terrestrial cryosphere. Production and delivery take place in a timely manner and are complemented by the constitution of long-term time series.

CLMS portfolio contains a number of global "Vegetation" biophysical products based upon the data acquired by the Ocean and Land Colour Instrument (OLCI) and the Sea and Land Surface Temperature Radiometer (SLSTR) onboard the Sentinel-3 (S3) platform. All various biophysical algorithms ingest the same surface reflectances derived from the OLCI and SLSTR Level 1B data by a harmonized pre-processing chain.

This document provides a detailed description of the CLMS OLCI and SLSTR surface reflectances 300m V2.3. It includes an overview of the retrieval methodology, and information on file naming, file format, product content, product characteristics, data policies and data access. It reports the main results of the quality assessment and validation of the product and makes recommendation for an efficient use of the product.

1 BACKGROUND OF THE DOCUMENT

1.1 SCOPE AND OBJECTIVES

The surface reflectance is the Top-Of-Canopy (TOC) reflectance derived from the Top Of Atmosphere (TOA) reflectance after correction for the temporally, spatially and spectrally varying scattering and absorbing effects of atmospheric components.

The Product User Manual (PUM) is the primary document that users have to read before handling the products. It gives an overview of the product characteristics, in terms of algorithm, technical characteristics, and main validation results. The key information is summarized in Table 1.

Table 1: Summary of key information

Input data Sentinel-3 OLCI and SLSTR Level 1B Full Resolution frames
Retrieval algorithm Pixel classification, co-registration between OLCI and SLSTR acquisitions, reprojection and resampling in a regular grid, and atmospheric correction.
File naming cgl_TOC_<TIME>_<AREA>_<SATELLITE>_v<version>.nc Example: cgl_TOC_20180721151742_X14Y00_S3B_v2.3.4.nc
File format and product content Multi-band netCDF file including surface reflectances and uncertainties for 15 OLCI spectral bands and 5 SLSTR spectral bands, observation and illumination angles, 4 flag layers
Projection and grid information EPSG:4326 with WGS 84 datum and grid resolution of 0.00297619°
Spatial information Longitude [180°W – 180°E], Latitude [~85°N – ~65°S]
Temporal information Acquisition time of the satellite sensor
Data policies

Free and open access
Data access Through the CLMS website: http://land.copernicus.eu/ .
Contact Through the CLMS website: https://land.copernicus.eu/en/contact-service-helpdesk

1.2 CONTENT OF THE DOCUMENT

This document is structured as follows:

- Chapter 2 presents the retrieval methodology
- Chapter 3 describes the technical characteristics of the product
- Chapter 4 reports the main results of the quality assessment
- Chapter 5 gives insights about product usage

The requirements are recalled in Annex.

1.3 RELATED DOCUMENTS

1.3.1 Applicable documents

AD1: Part 2: Technical specifications of Framework Service Contract – Operation of the bio-geophysical variables systematic monitoring of the Global Land Component of the Copernicus Land Service 'CGLOPS' JRC/2023/OP/0273, 19th April 2023.

Available at https://etendering.ted.europa.eu/cft/cft-display.html?cftId=13795

1.3.2 Input

Document ID	Descriptor
CGLOPS1_ATBD_S3-AC-V1.1	Algorithm Theoretical Basis Document of the Version 1.1 of the Atmospheric Corrections applied on Sentinel-3 OLCI and SLSTR data
CGLOPS1_VR_S3-TOC-Reflectance300m-V2.3	Quality assessment and validation report of the OLCI and SLSTR surface reflectance 300m Version 2.3 product

CGLOPS-1_QAR_S3-CloudMask	Report presenting the evaluation of performances of cloud, snow and shadow detection performed on the Sentinel-3 sensors data for the specific needs of global biophysical production
CGLOPS-1_VR_S3-TOCr300m-V1	Quality assessment and validation report of the OLCI and SLSTR surface reflectance 300m Version 1 product

Available at https://land.copernicus.eu/en/technical-library

1.3.3 External documents

Document ID	Descriptor
GCOS#245	The 2022 GCOS ECVs Requirements

https://library.wmo.int/records/item/58111-the-2022-gcos-ecvs-requirements-gcos-245?offset=1

IdePix_ATBD	Sentinel-3 OLCI IdePix Algorithm Theoretical Basis Document, V1.0, 4 May 2022

Available at https://zenodo.org/record/6517333

SLSTR_L1_ATBD	SLSTR ATBD for Level 1 observables – S3-TN-RAL-SL-032, I8.1, 1 April 2021

Available at https://sentiwiki.copernicus.eu/__attachments/1672112/S3_TN_RAL_SL_032_-Sentinel-3 SLSTR Level-1 Observables ATBD 2021 - 8.1.0.pdf

S3MPC.RAL.TN.020	Sentinel-3 SLSTR VIS and SWIR Channel Vicarious Calibration Adjustments, I3.0, 22 May 2021.

https://sentinels.copernicus.eu/documents/247904/4620074/Sentinel-3-SLSTR-VIS-and-SWIR-Channel-Vicarious-Calibration-Adjustments.pdf

S3 MRTD	Sentinel-3 Mission Requirements Traceability Document, I1.0, 7 February 2011

https://earth.esa.int/eogateway/documents/20142/1564943/Sentinel-3-Mission-Requirements-Traceability-Document-MRTD.pdf

S3.PN-OLCI-L1.12	Product Notice introducing the new Sentinel-3A and -3B OLCI Level-1B Collection 4

https://sentiwiki.copernicus.eu/__attachments/1672112/S3.PN-OLCI-L1.12_-_Sentinel_3_OLCI Product Notice Level 1B 2024 - 1.0.pdf

SLSTR Processing Baseline	Evolutions of the processing baseline used to process Sentinel-3 SLSTR data since July 2016

https://sentiwiki.copernicus.eu/web/slstr-processing

2 ALGORITHM

2.1 OVERVIEW

The Copernicus Global Land Operations (CGLOPS-1) processing lines generating biophysical products from Sentinel-3 OLCI and SLSTR data rely all on a common pre-processing chain which ingests L1B frames and delivers surface reflectances (Figure 1). This pre-processing chain is made of various modules which perform the pixel classification (Idepix), the co-registration between OLCI and SLSTR acquisitions, the reprojection and resampling in the CGLOPS-1 regular grid and the atmospheric corrections. Idepix is developed in the framework of the Sentinel-3 Optical Mission Performance Cluster (OPT-MPC) under ESA contracts and imbedded in the Sentinel Application Platform (SNAP). The co-registration, reprojection and resampling module and the Atmospheric Correction (AC) module are developed by CGLOPS-1.

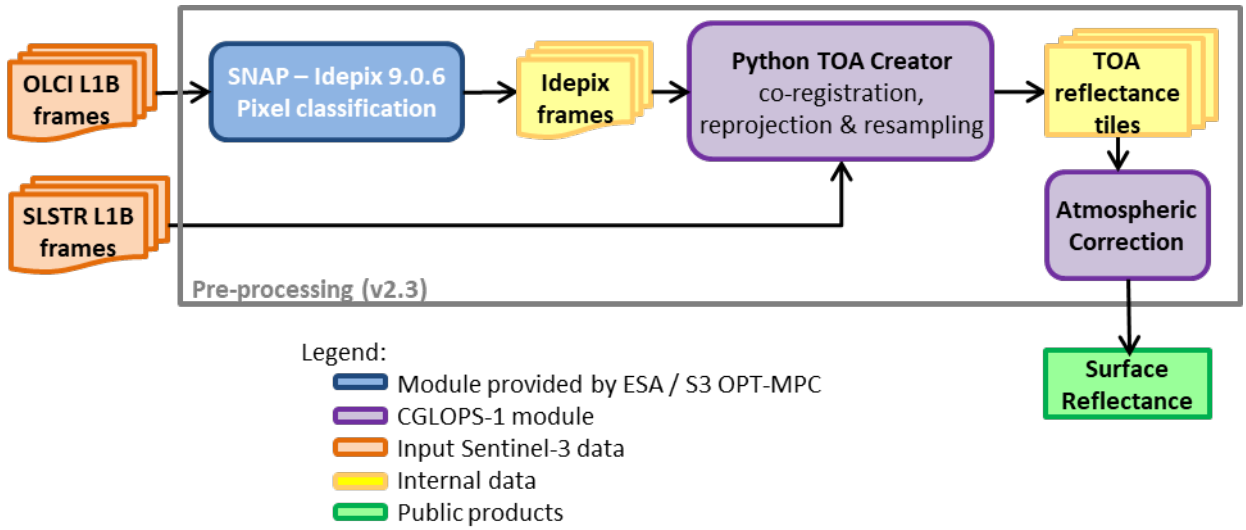


Figure 1: Pre-processing chain of the CGLOPS-1 Sentinel-3 based biophysical chains.

2.2 INPUT DATA

The historic (June 2018 – March 2025) surface reflectance V2.3 products are derived from the following L1B input data versions:

- For OLCI, the version OL__L1_.004.00.00 is used [S3.PN-OLCI-L1.12]. These OLCI data are accessible through the Copernicus Dataspace Ecosystem. An example of metadata dump of input OLCI L1 files is given in Annex 2.

- For SLSTR, the baseline collection 4 (SL__L1_.004) available at the Copernicus Dataspace Ecosystem, is used. The actual minor version depends on the year [SLSTR Processing Baseline]. Examples of metadata dump of input SLSTR L1 files are given in Annex 2.

The NRT (from 1st April 2025 onwards) surface reflectance V2.3 products are derived from the latest version of OLCI and SLSTR L1B data available at the Copernicus Dataspace Ecosystem.

For historic and NRT products, the used OLCI and SLSTR L1B source files can be found in the metadata of each surface reflectance file (Table 12).

Ancillary data for atmospheric corrections include the ozone and water vapor content, the aerosol content, the sea-level surface pressure, the surface temperature and the surface elevation. They are detailed in the ATBD [CGLOPS1_ATBD_S3-AC-V1.1].

2.3 RETRIEVAL METHODOLOGY

The surface reflectance refers to the portion of the incident energy reflected by the surface in a given spectral band and without atmospheric interferences.

The first step of the pre-processing chain (Figure 1) is the pixel classification which allows to identify clouds, snow and shadows. This step relies on separate schemes for OLCI and SLSTR input frames:

- For OLCI data, the pixel classification is performed by the IdePix processor (version 9.0.6). It is based on a neural network approach [IdePix_ATBD]. As input for the neural network, the square roots of the OLCI TOA reflectances (obtained from an internal radiance-to-reflectance conversion) in all 21 OLCI bands are used. As output, the neural network finally provides, per pixel, the pixel features as specified in the pixel_classif_flags (Table 10).
- For SLSTR data, the pixel classification is provided as Level 1B quality flag (cloud_an, Table 8). This latter results from a Basic Cloud Identification, based on a collection of separate cloud tests. The clouds are identified by using test over the brightness temperature data in the thermal channels and the reflectance data in the shortwave infrared and visible channels [SLSTR_L1_ATBD, Section 7.4.3].

The second step performs first the co-registration of the OLCI and SLSTR frames. An output reflectance product is generated only if a matching OLCI and SLSTR pair is available. This is consistent with the approach taken by ESA to produce the Sentinel-3 L2 SYN product. For the years 2018, 2019 and 2022, OLCI and SLSTR frame pairing is based on the cycle number, the relative orbit number and the frame along track coordinate, for S3A/S3B separately. Due to a faulty calculation of the relative orbit number of the SLSTR sensor in the original L1B data during the period May 2020-

April 2021, we changed to a more reliable start/stop time matching (checking for overlap) for years 2020, 2021 and from 2023 onwards, including the NRT operations. Note that, for the years 2018, 2019, 2022, applying this method provides the same results.

The co-registration approach relies on the per pixel latitude and longitude information available in the Level1B input files. Based on this information, the frames are reprojected and resampled in the regular latitude/longitude grid, with a spatial resolution of 1/336°, in the WGS84 projection. Projection to the regular grid is done through a Python reprojection module, using a nearest neighbor resampling based on scipy spatial cKDTree. First, a reprojection lookup table is created. Secondly, all related bands are reprojected using this LUT. Additionally, SLSTR angles bands are up-sampled and smoothed to align the resolution between all bands.

At the input of the AC module, some adjustment factors are applied on the TOA reflectances:

- A factor of 0.98 is applied on all OLCI-A spectral bands, except band 0a21, to reduce the brightness of OLCI-A acquisitions by 2% and make them more consistent with the OLCI-B data. This calibration adjustment is recommended by the S3 OPT-MPC and results from analysis of OLCI data acquired during the tandem phase (Clerc et al., 2020).
- Calibration adjustment coefficients are applied on SLSTR visible and SWIR bands. They are based upon the analysis of vicarious calibration results by several expert groups and are recommended by ESA [S3MPC.RAL.TN.020].

Finally, the atmospheric correction is performed [CGLOPS1_ATBD_S3-AC-V1.1]. The Simplified Method for the Atmospheric Correction (SMAC) (Rahman and Dedieu, 1994), based on the 6S radiative transfer formulation of the satellite signal (Vermote et al., 1997), is chosen as AC scheme due to its speed, robustness and analytical formulation. All the pertinent radiative quantities are parameterized as a function of auxiliary data: gas content (mainly ozone and water vapor), aerosol content, sea-level surface pressure and surface elevation that drives the molecular (Rayleigh) scattering. The aerosol model is the continental model. Necessary ancillary data come from different sources, in particular from the Copernicus Atmosphere Monitoring Service (CAMS) (Table 2). The atmospheric correction quality flag, so-called AC_process_flag (Table 11), gives information on how the atmospheric corrections are performed. It indicates the confidence on the atmospheric corrections results.

An uncertainty is associated to the resulting surface reflectance. It is calculated as the quadratic combination of individual uncertainty associated to TOA reflectance, aerosol optical thickness (AOT), ozone, water vapor, surface pressure and aerosol model. The first five individuals are calculated by the help of Jacobians thanks to the analytical basis of SMAC. The last one, the uncertainty on aerosol model, is statistically estimated. The uncertainty on TOA reflectance is included in the OLCI L1B input data. As the SLSTR L1B input data do not contain any unertaitny estimate, a relative uncertainty of 2% (0.02 in absolute) on TOA reflectances, as defined in the S3

SLSTR mission requirements [S3 MRTD], has been assigned. Details on uncertainty characterization and propagation are given in CGLOPS1_ATBD_S3-AC-V1.1.

Table 2: Sources of ancillary data for atmospheric correction, used for NRT production and historical off-line reprocessing

Ancillary data	NRT production	Historical re-processing
Aerosol optical thickness	CAMS NRT product	CAMS re-analysis
Ozone total column	CAMS NRT product	CAMS re-analysis
Water vapor column	OLCI L1B input data	
Surface pressure	OLCI L1B input data	
Surface temperature	Climatology based upon MERRA-2 monthly mean of 2018	

2.4 LIMITATIONS

The surface reflectance product is only generated when a matching OLCI/SLSTR pair is available. This means that, if one of the two sensors has a missing L1B product for that point in time, the surface reflectance product is not created.

The pixel classification by IdePix suffers from some well-known limitations which are confirmed by the CGLOPS-1 specific analysis [CGLOPS-1_QAR_S3-CloudMask]. Idepix tends to omit thin or semi-transparent clouds and cloud edges. In some cases, IdePix over-detects clouds over bright surfaces, such as bare areas and, less frequently, water shorelines. Furthermore, the snow mask of IdePix is not fully adequate: clear sky snow pixels are sometimes incorrectly masked as clouds. For most of the downstream CGLOPS-1 vegetation variables, this makes no difference as pixels contaminated by clouds, cloud shadow or snow are excluded from the Leaf Area Index (LAI), Fraction of Absorbed Photosynthetically Active Radiation (FAPAR), Dry Matter Productivity (DMP), Net Primary Production (NPP) and Fraction of green Vegetation Cover (FCover) processing. However, it is an important artefact for algorithms that require adequate snow detection (e.g. Bidirectional Reflectance Distribution Function (BRDF) correction for the Normalized Difference Vegetation Index (NDVI) product).

The SLSTR cloud flag performs well to identify clouds but over-detects the thin cirrus. The thin cirrus mask is therefore not used on the Burnt Areas retrieval algorithm, as it severely limits the number of observations of the land surface.

The reprojection and resampling method creates some gaps at high latitudes. Indeed, for every pixel in the target 1/336° grid, the tool identifies the nearest pixel in the frames, up to a distance of 2/336°

(~=0.006°). At the edge of the swath (where viewing angles are the largest) and at higher latitudes (where the regular lat/lon grid is more "stretched"), this causes gaps in the resulting image, because the actual ground sampling distance of OLCI is larger than 0.006°. At high latitudes, the effect is slightly more pronounced for SLSTR bands, related to the larger ground sampling distance.

The SMAC algorithm is a parameterization of the radiative transfer model 6S (Vermote et al., 1997). 6S commonly agreed accuracy is 1%. SMAC approximation to 6S is also claimed to be within 1% for most situations under the assumption of lambertian surface (e. g. isotropic). However, for anisotropic surfaces, the SMAC approximation could be quite inaccurate to several %, especially for high atmospheric turbidity. Neither adjacency effects nor terrain slope correction is applied. These effects are, however, secondary at the spatial scale of 300 meters. Approximations in radiative transfer lead essentially to biases, depending on surface, angles and atmospheric content, and are certainly not a source of random noise. This source of uncertainty is thus not yet included in the uncertainty propagation model.

The method to assess the uncertainty due to aerosol model suffers some limitations that are detailed in CGLOPS1_ATBD_S3-AC-V1.1.

2.5 DIFFERENCES WITH PREVIOUS VERSIONS

The differences in the various pre-processing steps with the previous surface reflectance versions are detailed in the Table 3. The versions 1.0 and 1.1 of surface reflectances were the versions used as input to generate the NRT S3-based biophysical products. The version 1.1 was developed to correct the geolocation issue that affected the version 1.0 up to July the 1st 2022. This error was caused by an erroneous setting in the input reader attributes of SNAP 7.0.3, resulting in using tie points as basis for projection. Tie points provide a faster but less accurate geolocation compared to using pixel-based geocoding, listing for each pixel the corresponding coordinates. The version 1.1 was deployed in NRT operations from July 2022 until 28th February 2025. The timeline of the successive versions of OLCI and SLSTR L1B input data used to calculate the surface reflectances version 1.0 and version 1.1 can be found in https://operations.dashboard.copernicus.eu/processors-viewer.html.

The versions 2.x benefit from an in-house module for reprojection and resampling replacing the L1CSYN tool. This TOA creator tool not only improves the co-registration but also decreases the processing time compared with the use of the SNAP reprojection module and requires less storage. Versions 2.1 and 2.2 were intermediate releases which have not been used for published products. Version 2.3 uses the IdePix v9.0.6 and applies also the calibration correction coefficients on SLSTR and OLCI-A as recommended by ESA. Finally, the assessment of the uncertainties on surface reflectances is more accurate, considering the contribution of the aerosol model.

Table 3: Successive surface reflectance versions and their differences

Version	Differences	Status
V1.0	First version of surface reflectance products: • Based on L1CSYN 1.0.1 frames ○ SNAP 7.0.3 ○ L1CSYN 2.5 ○ Idepix 7.0.5 • Based on TOA reflectances v1.0.1 10°x10° tiles ○ SNAP 7.0.3 Reproject module • Atmospheric correction v1.0: uncertainty from aerosol model not accounted in the surface reflectance uncertainty estimate	Internal. Used as input for NRT production of S3-based biophysical product up to 30th June 2022.
V1.1	Update for correcting the geolocalization issue: • Based on L1CSYN 1.0.2 frames ○ SNAP 7.0.3 with geopixelEncoding ○ L1CSYN 2.5 ○ Idepix 7.0.5 • Based on TOA reflectances v1.0.2 10°x10° tiles ○ py-extract tool: reprojection and resampling • Atmospheric correction v1.0: uncertainty from aerosol model not accounted for the surface reflectance uncertainty estimate	Internal Used as input for NRT production of S3-based biophysical products from 1st July 2022 to 28th February 2025.
V2.3	Surface reflectance version 2.3 products: • Idepix 9.0.6 for pixel classification • Python TOA creator: (improved) co-registration, reprojection and resampling • Sentinel-3 OLCI-A calibration correction factor applied • Sentinel-3 SLSTR calibration adjustment coefficients applied • No_data values of SLSTR bands are not processed as valid data. • Atmospheric correction v1.1: uncertainty from aerosol model accounted for the surface reflectance uncertainty estimate	Public From June 2018 to the present. Used as input for NRT production of S3-based biophysical products from 1st March 2025.

3 PRODUCT DESCRIPTION

3.1 FILE NAMING

The Sentinel-3 surface reflectance 300m product follows the naming standard:

cgl_<Acronym>_<TIME>_<AREA>_<SATELLITE>_V<VERSION>.nc

where

- <Acronym> identifies the product. Here, "TOC" refers to the Top-Of-Canopy reflectance at 300m resolution
- <TIME> indicates the product reference date. It is expressed in YYYYMMDDHHmmss where YYYY, MM, DD, HH, mm and ss denote the year, the month, the day, the hour, the minutes, and the seconds, respectively.
- <AREA> gives the spatial coverage of the file. The surface reflectance products are distributed in 10°x10° tiles (3360 x 3360 pixels) named XxxYyy like in Figure 2.
- <SATELLITE> gives the platform name i.e. S3A or S3B
- <VERSION> shows the processing line version used to generate the surface reflectance products. The version is denoted as M.m.r (e.g. 2.0.1), with 'M' representing the major version (e.g. V2), 'm' the minor version (starting from 0) and 'r' the production run number (starting from 1) (Table 4).

Note: For the historical surface reflectance products, covering the period June 2018 – March 2025, the product version is V2.3.4. From the 1st April 2025 onwards, the product version is V2.3.1. This rollback of the run digit was necessary due to some internal necessity of the dissemination platform, known after the end of the processing of the historical products. However, the processing chain running V2.3.4 and V2.3.1 is strictly the same.

Table 4: Explanation in version numbering and recommendations for using efficiently the products

Versions	Differences	Recommendations
Major	Significant change to the algorithm.	Do not mix various major versions in the same application, unless it is otherwise stated.
Minor	Minor changes in the algorithm	Can be mixed in the same application, but require attention or modest modifications
Run	When a reprocessing is necessary to fix bugs and minor issues. The latter run automatically replaces the former.	Consider it as a drop-in replacement

3.2 FILE FORMAT

The surface Reflectances 300m Version 2.3 product is delivered as a multiband Network Common Data Form version 4 (netCDF4) raster file with metadata attributes compliant with version 1.6 of the Climate & Forecast conventions (CF V1.6). The netCDF file contains 52 data layers (Table 5).

Table 5: Layers of the surface Reflectances product files

Parameter	Description
Oa*xx*_toc (for *xx* in 02-12, 16-18, 21)	Surface reflectances for OLCI bands
Oa*xx*_toc_error (for *xx* in 02-12, 16-18, 21)	Uncertainty on surface reflectances for OLCI bands
S*x*_an_toc (for *x* in 1-3, 5-6)	Surface reflectances for SLSTR bands (A stripe, nadir view)
Sx_an_toc_error (for *x* in 1-3, 5-6)	Uncertainty on surface reflectances for SLSTR bands
SAA_*sensor* (for *sensor* in olci, slstr)	Solar azimuth angle for OLCI and SLSTR
SZA_*sensor* (for *sensor* in olci, slstr)	Solar zenith angle for OLCI and SLSTR
VAA_*sensor* (for *sensor* in olci, slstr)	Viewing azimuth angle for OLCI and SLSTR
VZA_*sensor* (for *sensor* in olci, slstr)	Viewing zenith angle for OLCI and SLSTR
cloud_an	Cloud flag from SLSTR level 1B input data
quality_flags	Quality flags from OLCI level 1B input data
pixel_classif_flags	Output from IdePix
AC_process_flag	Quality flag of the atmospheric correction processing

3.3 PRODUCT CONTENT

As indicated in Table 5, the surface reflectances data files contain 20 spectral reflectances and their associated uncertainties, 8 observation and illumination angles and 4 flags.

3.3.1 Reflectance bands

All reflectance bands ("Oa*_toc" for OLCI and "S*_an_toc" for SLSTR) and their associated uncertaintiy bands ("Oa*_toc_error" for OLCI and "S*_an_toc_error" for SLSTR) are stored as 16-

bit integer values. The digital number (DN) value for no data is 32767. No data in the spectral surface reflectances results from missing data in the input OLCI or SLSTR L1B files.

The physical range and scaling factors of reflectances and their uncertainties are given in Table 6. The physical values are retrieved from the DN by:

$$PhysicalValue = DN * Scaling_Factor + Add_Offset \quad \textbf{Eq. 1}$$

Table 6: Physical and DN ranges and scaling factors for reflectances and their uncertainties.

Type	Physical range	DN range	Scaling_Factor	Add_Offset
toc	[-1, 1]	[-20000, 20000]	0.00005	0
toc_error	[-1, 1]	[-20000, 20000]	0.00005	0

3.3.2 Observation and illumination angles

The observation and illumination angle bands include:

- VZA: Viewing Zenith Angle
- VAA: Viewing Azimuth Angle
- SZA: Solar Zenith Angle
- SAA: Solar Azimuth Angle

for OLCI ("*_olci") and SLSTR ("*_slstr"), respectively. The difference in scanning systems (push-broom for OLCI vs. conical scanner for SLSTR) results in different sampling geometry and non-simultaneous acquisitions, causing illumination and viewing angles to be (slightly) different. The viewing azimuth angle is ranging between -180° and 180° for OLCI (negative angles for east-looking, positive angles for west-looking), and between 0° and 360° for SLSTR (> 180° for east-looking, < 180° for west-looking) (Figure 2). Sun azimuth angles are generally positive and < 180°, as Sentinel-3 has a morning overpass and imagery is captured when the Sun is positioned in the eastern sky.

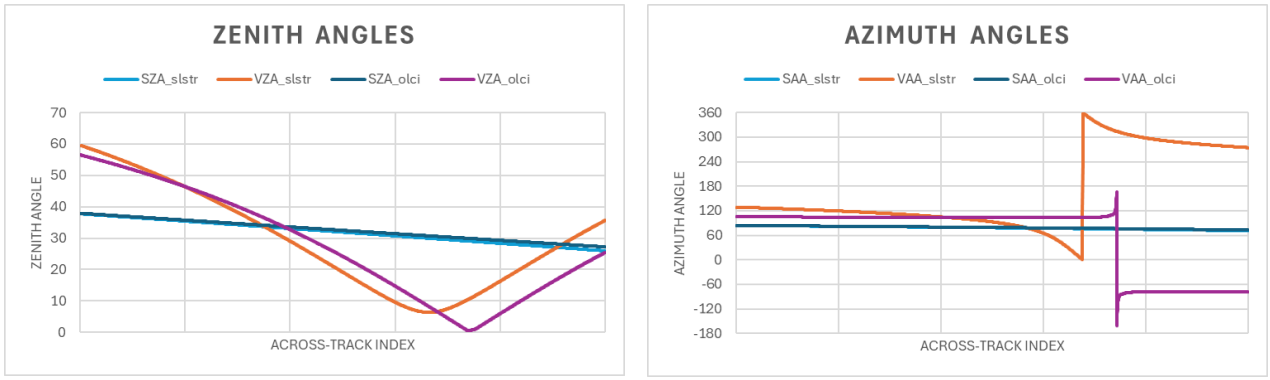


Figure 2: Comparison of across-track evolution of zenith angles (left) and azimuth angles (right) between OLCI and SLSTR

Zenith angles and OLCI azimuth angle information are stored as 16-bit integer values. The DN value for no data is 32767. The physical range and scaling factors are given in Table 7.

Note: the SLSTR azimuth angles are not stored as signed 16-bit integer values having a physical range of [-180°, 180°], as specified in the metadata of the netCDF files, but as unsigned 16-bit integer having a physical range of [0 - 360°] (Table 7). Consequently, the VAA_slstr and SAA_slstr bands need special attention when reading in (see section 5.3 for guidelines).

Table 7: Physical and DN ranges and scaling factors for angles

Type	Physical range	DN range	Scaling_Factor	Add_Offset
Azimuth angles OLCI	[-180°, 180°]	[-18000, 18000]	0.01	0
Azimuth angles SLSTR	[0, 360]	[0, 36000]	0.01	0
Zenith angles	[0°, 180°]	[0, 18000]	0.01	0

3.3.3 Annotation flags

The Sentinel-3 surface reflectance data files contain 4 quality layers:

- cloud_an: cloud flags from SLSTR Level 1B, stored as 16-bit integer values (Table 8). See 5.2.1 for more information.
- quality_flags: quality flags from OLCI Level 1B, stored as 32-bit integer values (Table 9). See 5.2.2 for more information.
- pixel_classif_flags: output of IdePix applied on OLCI data, stored as 16-bit integer values (Table 10).
- AC_process_flag: output of atmospheric correction module, as defined in the ATBD of Atmospheric Corrections [CGLOPS1_ATBD_S3-AC-V1], stored as one-byte values (Table 11).

Table 8: cloud_an flag encoding

Bit	Name	Value
0	Visible channels cloud test	1
1	1.37 µm threshold test	2
2	1.6 µm small-scale histogram test	4
3	1.6 µm large-scale histogram test	8
4	2.25 µm small-scale histogram test	16
5	2.25 µm large-scale histogram test	32
6	11 µm spatial coherence test	64
7	12 µm gross cloud test	128
8	11 µm/12 µm thin cirrus test	256
9	3.7 µm/12 µm medium/high level test	512
10	11 µm/3.7 µm fog/low stratus test	1024
11	11 µm/12 µm view difference test (uses both views)	2048
12	3.7 µm /11 µm view difference test (uses both views)	4096
13	11 µm/12 µm thermal histogram test	8192
	DN no data value	-1

Table 9: quality_flags encoding

Byte/Bit	Name	Value
0/0	saturated_Oa21	1
0/1	saturated_Oa20	2
0/2	saturated_Oa19	4
0/3	saturated_Oa18	8
0/4	saturated_Oa17	16
0/5	saturated_Oa16	32
0/6	saturated_Oa15	64
0/7	saturated_Oa14	128
1/0	saturated_Oa13	256
1/1	saturated_Oa12	512
1/2	saturated_Oa11	1024
1/3	saturated_Oa10	2048
1/4	saturated_Oa09	4096
1/5	saturated_Oa08	8192
1/6	saturated_Oa07	16384
1/7	saturated_Oa06	32768
2/0	saturated_Oa05	65536
2/1	saturated_Oa04	131072
2/2	saturated_Oa03	262144
2/3	saturated_Oa02	524288
2/4	saturated_Oa01	1048576
2/5	Dubious	2097152
2/6	sun_glint_risk	4194304
2/7	Duplicated	8388608
3/0	Cosmetic	16777216
3/1	Invalid	33554432
3/2	straylight_risk	67108864
3/3	Bright	134217728
3/4	tidal_region	268435456
3/5	fresh_inland_water	536870912
3/6	Coastline	1073741824
3/7	Land	-2147483648

Table 10: pixel_classif_flags encoding

Bit	Name	Value
0	IDEPIX_INVALID	1
1	IDEPIX_CLOUD	2
2	IDEPIX_CLOUD_AMBIGUOUS	4
3	IDEPIX_CLOUD_SURE	8
4	IDEPIX_CLOUD_BUFFER	16
5	IDEPIX_CLOUD_SHADOW	32
6	IDEPIX_SNOW_ICE	64
7	IDEPIX_BRIGHT	128
8	IDEPIX_WHITE	256
9	IDEPIX_COASTLINE	512
10	IDEPIX_LAND	1024
11	IDEPIX_MOUNTAIN_SHADOW	2048
	DN no data value	-1

Note that these flags are **not** mutual exclusive and that multiple combinations are possible. For instance, a pixel can have the value 192: IDEPIX_SNOW_ICE (64) and IDEPIX_BRIGHT (128).

The IDEPIX_LAND flag, in the IdePix version 9.0.6 (see Section 2.3), excludes inland waters. This was also the case in surface reflectance V1 (Table 3).

Table 11: AC_process_flag encoding

Bit	Name	Flag value
0	Reserved – not implemented	0 fixed
2,1	Bit 2: 0, Bit 1: 0 Nominal AOT value (AOT ≤0.5): High confidence level	0
	Bit 2: 0, Bit 1: 1 High AOT value (0.5 < AOT ≤1.0): Moderate confidence level	2
	Bit 2:1, Bit 1: 0 Very high AOT value (1.0 < AOT ≤1.5): Low confidence level	4
	Bit 2:1, Bit 1: 1 Critical AOT value (1.5 < AOT): Very low confidence level	6
3	High solar zenith angle (SZA>65°)[1]: Low confidence level	8
4	High viewing zenith angle (VZA>65°)[2]: Low confidence level	16
5	Use of a climatology (This flag value indicates the use, as backup, of the MERRA-2 climatology in case of missing CAMS NRT ancillary data)	32
6	Not used	64
7	Sign bit	N.A.
	DN no data value	-1

3.3.4 Metadata

The netCDF files contain several metadata attributes:

- globally, on the level of the entire file (Table 12);
- at the level of the data variable (Table 13 gives examples for the *Oa02_toc* variable);
- at the level of the coordinate variables for latitude ('lat') and longitude ('lon') (Table 14);
- at the level of the grid mapping (or coordinate transformation) variable ('crs') (Table 15).

[1] Since only SLSTR nadir images are used, the SZA and VZA of OLCI and SLSTR are very close.
[2] Note that a VZA > 65° is never reached on OLCI and SLSTR nadir images.

Table 12: Global attributes in the surface reflectances netCDF files

Attribute name	Description	Example(s)
conventions	Version of the CF-Conventions used	CF-1.6
archive_facility	Specifies the name of the institution that archives the product	CDSE
copyright	Text to be used by users when referring to the data source of this product in publications (copyright notice)	Copernicus Land Monitoring Service
history	A global attribute for an audit trail. One line, including date in ISO-8601 format, for each invocation of a program that has modified the dataset.	2025-04-01 S3 pre-formatter s3_TOA_creator v0.0.64 2025-04-01 SMACCL v2.3.1 - kernel 1.01.00
identifier	Unique identifier for the product	urn:cgl:sen3_l2_toc_333m:cgl_TOC_20250401114425_X17Y01_S3A_v2.3.1
institution	The name of the institution that produced the product	vito
orbit_type	Orbit type of the orbiting platform(s)	LEO
parent_identifier	Identifier of the product (time series) in the Copernicus Global Land Service metadata catalogue.	urn:cgl:sen3_l2_toc_333m
platform	Name(s) of the orbiting platform(s)	S3A,S3B
processing_level	Product processing level	L2
processing_mode	Processing mode used when generating the product	Offline
product_version	Version of the product (VM.m.r)	2.3.4
sensor	Name(s) of the sensor(s) used	OLCI, SLSTR
source	The method of production of the original data	Derived from EO satellite imagery
title	A description of the contents of the file	Sentinel 3A L2 Top Of Canopy Tile X30Y12 at 333m resolution
comment	Free field where we put the names of the used OLCI/SLSTR products	source: S3A_OL_1_EFR____20191011T142429_20191011T142729_20240601T094335_0179_050_167_3780_MAR_R_NT_004.SEN3, S3A_SL_1_RBT____20191011T142429_20191011T142729_20210116T142958_0179_050_167_3780_LR1_R_NT_004.SEN3

Table 13: Description of netCDF attributes for the *Oa02_toc* variable

Attribute	Description	Example values for *Oa02_toc* reflectance
long_name	A descriptive name that indicates a variable's content. This name is not standardized. Required when a standard name is not available.	TOC reflectance for OLCI acquisition band Oa02
scale_factor	Multiplication factor for the variable's contents that must be applied in order to obtain the real values. Omit in case the scale is 1.	5.e-05
units	Units of the variable's content, taken from Unidata's udunits library.	1
valid_range	Smallest and largest values for the variable. Missing data is to be represented by one or several values outside of this range.	-20000, 20000
_FillValue	Single value used to represent missing or undefined data and to pre-fill memory space in case a non-written part of data is read back. Value must be outside of valid_range.	32767
missing_value	Single value used to represent missing or undefined data, for applications following older versions of the standards. Value must be outside of valid_range.	32767
grid_mapping	Reference to the grid mapping variable	crs

Table 14: Description of netCDF attributes for coordinate variables (latitudes and longitudes)

Attribute	Description	Example
class	Dataset type	dimension_scale
dimension_labels	Identifies the corresponding dimension	lon
standard_name	A standardized name that references a description of a variable's content in CF-Convention's standard names table. Note that each standard_name has corresponding unit (from Unidata's udunits).	longitude
long_name	A descriptive name that indicates a variable's content. This name is not standardized. Used when a standard name is not available (FDOB layers).	longitude
units	Units of the variable's content, taken from Unidata's udunits library.	degrees_east
axis	Identifies latitude, longitude, vertical, or time axes.	x
_CoordinateAxisType	This specifies that the variable is a space or time coordinate variable, such as lat, lon, altitude or time.	lon

Table 15: Description of netCDF attributes for the grid mapping variable

Attribute	Description	Example
GeoTransform	Six coefficients for the affine transformation from pixel/line space to coordinate space, as defined in GDAL's GeoTransform	120.0000000000 0.0089285714 0.0 -35.0000000000 0.0 -0.0089285714
_CoordinateAxisTypes	This Coordinate Transform applies to any Coordinate System with the specified list of Axis types.	GeoX GeoY
_CoordinateTransformType	This indicates that the variable is a Coordinate Transform Variable	Projection
grid_mapping_name	Name used to identify the grid mapping	latitude_longitude
inverse_flattening	Used to specify the inverse flattening (1/f) of the ellipsoidal figure associated with the geodetic datum and used to approximate the shape of the Earth	298.257223563
long_name	A descriptive name that indicates a variable's content.	coordinate reference system
longitude_of_prime_meridian	Specifies the longitude, with respect to Greenwich, of the prime meridian associated with the geodetic datum	0.0
semi_major_axis	Specifies the length, in metres, of the semi-major axis of the ellipsoidal figure associated with the geodetic datum and used to approximate the shape of the Earth	6378137.0
spatial_ref	Spatial reference system in OGC's Well-Known Text (WKT) format	GEOGCS["WGS 84",DATUM["WGS_1984",…AUTHORITY["EPSG","4326"]]

3.4 PRODUCT CHARACTERISTICS

3.4.1 Projection and Grid Information

The surface reflectance product is displayed in a regular latitude/longitude grid using the World Geodetic System 1984 (WSG 84) projection as defined by the following Well-Known Text (WKT) format:

```
GEOGCS["WGS 84",
       DATUM["WGS_1984",
             PHEROID["WGS 84",
```

```
                6378137,
                298.257223563,
                AUTHORITY["EPSG","7030"]],
        TOWGS84[0,0,0,0,0,0,0],
        AUTHORITY["EPSG","6326"]],
PRIMEM["Greenwich",
        0,
        AUTHORITY["EPSG","8901"]],
UNIT["degree",
      0.0174532925199433,
      AUTHORITY["EPSG","9108"]],
AUTHORITY["EPSG","4326"]]
```

The resolution of the grid is (1/336)°. Pixel tie point coordinates are at the centre of the pixel. The bounding box coordinates of the area covered by the pixel (e.g. to be used by GDAL) can be calculated as:

$$Y_{upper} = Y_{center\ pixel} + {}^{0.5}/_{336} \text{ and } Y_{lower} = Y_{center\ pixel} - {}^{0.5}/_{336} \quad \textbf{Eq. 2}$$

$$X_{left} = X_{center\ pixel} - {}^{0.5}/_{336} \text{ and } X_{right} = X_{center\ pixel} + {}^{0.5}/_{336} \quad \textbf{Eq. 3}$$

The upper left tie point of the grid is 85°N,180°W (center pixel coordinates) and covers up to 65°S,180°E (center pixel coordinates).

OLCI and SLSTR do not have exactly the same footprint for Sentinel 3 input frames with the same <time> in the filename:

(1) in the across-track direction, the SLSTR (nadir) footprint is slightly larger, related to the different swath (1270 km for OLCI, 1400 km for nadir view SLSTR). Note that OLCI and SLSTR nadir are asymmetrical with respect to the nadir point. Indeed, OLCI is tilted westwards with 12.6° to mitigate the negative impact of sun glint contamination for water remote sensing.

(2) because of slight acquisition time difference of OLCI and SLSTR, frames with the same <time> (start of acquisitions) have a slightly different footprint in the along-track direction.

3.4.2 Spatial Information

The OLCI and SLSTR surface reflectances cover the globe from longitude 180°E to 180° W and from latitude ~85°N to ~65°S according to the 10°x10° tiling shown in Figure 3. Since the grid starts at 85°N, the equator is actually located in the middle of Y08 tiles.

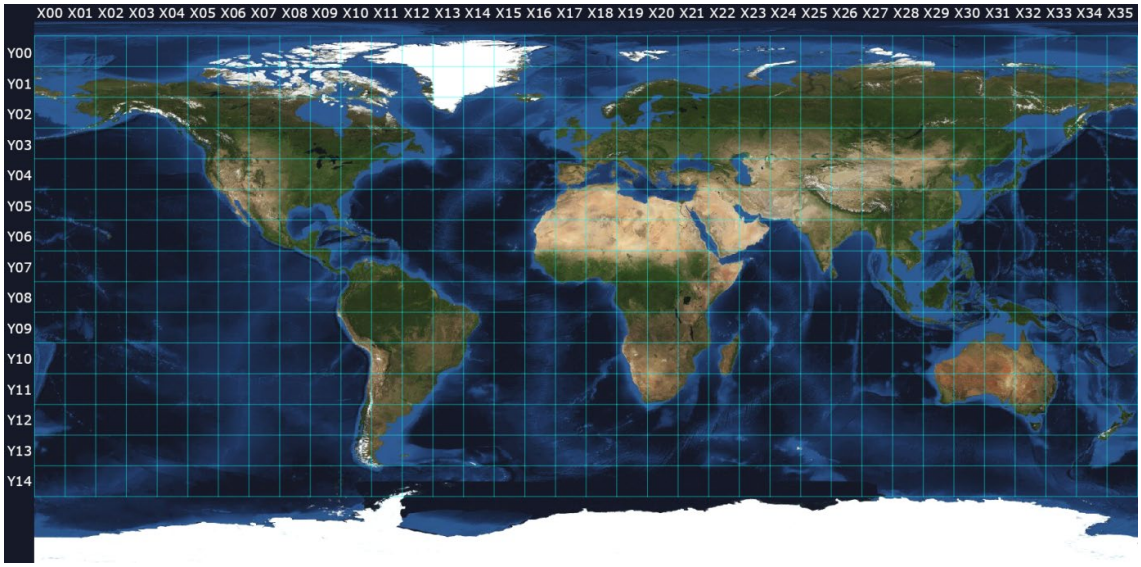


Figure 3: Grid tiles on world map[3]

3.4.3 Temporal Information

The start acquisition time of OLCI and SLSTR sensors can sometimes be slightly different. However, the date, as indicated in the filename (Section 3.1), of the surface reflectance products is based on the start of acquisition time of OLCI frames. This design choice is because the OLCI sensor has a narrower time window than the SLSTR sensor and, thus, defines the range of products to be produced.

The Sentinel-3 OLCI and SLSTR surface reflectance V2.3 dataset covers the period from June 2018 to the present, i.e. the period when both S3A and S3B satellites are in operations.

3.5 Data Policies

The Copernicus component is governed by Regulation (EU) No 2021/696 of the European Parliament and of the Council of 28 April 2021 establishing the Union Space Programme and the European Union Agency for the Space Programme.

The Copernicus land monitoring products and services are made available on a principle of **full, open and free access**, as established by the Commission Delegated Regulation (EU) No 1159/2013 of 12 July 2013.

How to cite and reference the products and services of the Copernicus Land Monitoring Service is explained here: https://land.copernicus.eu/en/data-policy.

[3]Background image source: http://server.arcgisonline.com/arcgis/rest/services/ESRI_Imagery_World_2D/MapServer

3.6 ACCESS AND CONTACTS

The Sentinel-3 surface reflectance V2.3 products are available through the Copernicus Land Monitoring Service website at the address: https://land.copernicus.eu/ with contact information (helpdesk) on https://land.copernicus.eu/en/contact-service-helpdesk.

4 QUALITY ASSESSMENT

The validation of CLMS Sentinel-3 surface reflectance V2.3 products derived from OLCI and SLSTR was performed for the period from July 2018 to December 2023 [CGLOPS1_VR_S3-TOC-Reflectance300m-V2.3]. The methodology is based on two main validation approaches: direct validation (i.e., comparison of satellite products with in-situ measurements) and indirect validation (or product intercomparison). The various criteria of performance and metrics were assessed: completeness, spatial consistency, temporal consistency and error evaluation. The product intercomparison was performed using Sentinel-3 ESA L2 SYN products, whereas the direct validation was evaluated against ground data from CEOS RadCalNet and complemented with ESA Fiducial Reference Measurements (FRM) and limited dataset of HYPERNETS project (https://hypernets.eu). Moreover, the consistency of surface reflectance V2.3 derived from different satellites (Sentinel-3A versus Sentinel-3B during the tandem phase) and different sensors (OLCI versus SLSTR similar spectral channels) was evaluated.

Main conclusions for each criterion are:

- **Product content**

The visual inspection of global maps at reduced resolution and selected tiles and Areas of Interest (AOI) at full resolution shows consistent distribution of reflectance values, associated uncertainties (typically ranging between 2% and 6%, except for blue spectral domain), illumination and viewing angle values.

Regarding the use of ancillary quality layers for OLCI and SLSTR (Section 3.3.3), the pixel_classif_flags (Table 10) is the most effective to remove clouds but with two main limitations: i) conservative approach and, ii) misidentification of clouds and snow. Cloud_an (Table 8) systematically removes observations for northern latitudes in winter time, whilst clouds are not properly removed in the equatorial belt. The border of the clouds is also not well detected. Consequently, in addition to cloud_an, it is recommended the use pixel_classif_flags to remove several outliers and cloud contamination. Few pixels are removed when quality_flags (Table 9) and AC_process_flag (Table 11) are used, except for the bit 3 of AC_process_flag (SZA > 65º) which discard large areas in northern and southern latitudes during wintertime.

- **Product completeness of snow-free best quality pixels**

Excluding pixels with SZA larger than 65° creates long data gaps over northern latitudes and southern latitudes during wintertime. On the other hand, SLSTR surface reflectances provide higher number of valid good quality observations than OLCI surface reflectances, due to a less restrictive quality flags (Section 3.3.3).

- **Temporal consistency**

CLMS Sentinel-3 surface reflectances V2.3 are consistent with ESA L2 SYN and RadCalNet measurements reproducing properly the temporal variations. CLMS OLCI and SLSTR reflectances display similar temporal trends for similar spectral channels.

- **Direct validation**

The direct validation of surface reflectance V2.3 retrievals close to nadir (VZA < 10º) show a good accuracy in comparison with CEOS RadCalNet multi-temporal measurement from 4 sites (desert and sparse vegetation). Indeed, the mean bias (B) is typically within ±2% for most OLCI channels (slightly large bias for Oa02 and Oa21 bands). The overall uncertainty (quantified by the Root Mean Square Deviation (RMSD)) is around 7-9% with around 33-46% of pixels meeting the goal requirement (i.e. 3%, Section 7.1) and 55-65% of pixels within the threshold level (5%, Section 7.1). The mean bias is within ±1.5% for all SLSTR spectral channels. The overall uncertainty is around 7-10% with around 30-37% of pixels meeting the goal requirement and 48-56% of pixels within the threshold level. These are improved results compared to the surface reflectance V1 [CGLOPS-1_QAR_S3-TOCr300m-V1] (for OLCI, B around ±3% and RMSD around 9-13%; for SLSTR, B up to 2% for VNIR and worse than -10% for SWIR). CLMS surface reflectance V2.3 also performs significantly better than ESA L2 SYN (OLCI: B around 3-10% and RMSD around 11-15%; SLSTR: B > 6% for VNIR and up to -12% for SWIR with RMSD larger than 12%).

The CLMS surface reflectances V2.3 agree well with in-situ data from other initiatives. In particular, they display similar spectral signatures than ESA FRMs. In comparison with HYPERNETS multi-angular measurements from 3 sites (desert, agricultural and deciduous broadleaved forest), the median deviation is within -1% to 6%.

- **Product intercomparison**

The comparison between surface reflectance V2.3 from Sentinel-3A versus Sentinel-3B over desert calibration sites during the tandem phase shows very good agreement: R>0.98 and bias within ±0.9% for OLCI and within ±1.5% for SLSTR.

The CLMS surface reflectance V2.3 from OLCI and SLSTR equivalent channels over desert calibration sites shows remarkably good consistency for red (OLCI Oa08 vs. SLSTR S2, with mean bias within ±0.3%) and NIR (OLCI Oa17 vs. SLSTR S3, with mean bias within ±0.8%). Green OLCI Oa06 spectral channels shows higher values than SLSTR S1 (mean bias < 4%). OLCI and SLSTR reflectances show a good agreement over various AOIs covering different biome types (R > 0.85 for most cases). This agreement is increased compared to the version V1 thanks to an improved OLCI-SLSTR co-registration.

The CLMS Sentinel-3 surface reflectance V2.3 are compared with ESA L2 SYN over different AOIs. Both products display very similar spatial patterns. For visible OLCI and SLSTR spectral channels, CLMS products generally provides lower surface reflectance values than ESA products. The exceptions are the blue OLCI bands where a large positive mean bias (i.e. CLMS > ESA) is found over some AOIs. For NIR OLCI and SLSTR spectral channels, slightly negative or close to zero mean bias is obtained. At the opposite, a positive mean bias is obtained for SLSTR SWIR channels. It is worth to note that the CLMS and ESA products agree well over the snow and ice area in the visible and NIR bands.

Concluding remarks

In summary, CLMS surface reflectance V2.3 products are reliable for public dissemination and to generate biophysical products. Several improvements are noticed respect to surface reflectance V1:

- Optimal agreement between OLCI-A and OLCI-B and more realistic reflectance values for SLSTR S5 and S6 channels thanks to the application of ESA calibration adjustment factors.
- Improved agreement between OLCI and SLSTR reflectances due to a better co-registration.
- Improved accuracy in the comparison with RadCalNet measurements, partly attributed to the application of ESA calibration adjustment factors for OLCI-A and SLSTR SWIR bands
- Slightly improved cloud and snow identification due to updated IdePix processor.

However, some limitations remain: i) conservative cloud screening approach of IdePix Pixel_classif_flags; ii) misidentification of snow targets as clouds and iii) large fraction of missing data during wintertime when discarding pixels with SZA larger than 65°, where the atmospheric correction is not optimal (using or not this flag is left at each user's judgement).

5 PRODUCT USAGE

5.1 RECOMMENDATION ON USING THE ANNOTATION FLAGS

It is recommended to use the annotation flags (Section 3.3.3) to discard surface reflectances:

- In case of OLCI-related reflectance bands:
  - quality_flags layer (see Table 9):
    - 'saturated_Oa*', depending on which band the correction is applied to, exclude when raised
    - 'fresh_inland_water', exclude when raised
  - pixel_classif_flags (see Table 10):
    - IDEPIX_INVALID: exclude when raised
    - IDEPIX_LAND: include when raised
    - IDEPIX_CLOUD: exclude when raised
    - IDEPIX_CLOUD_AMBIGUOUS: exclude when raised
    - IDEPIX_CLOUD_BUFFER: exclude when raised
    - IDEPIX_CLOUD_SHADOW: exclude when raised
    - IDEPIX_SNOW_ICE: exclude when raised
    - IDEPIX_MOUNTAIN_SHADOW: depending on the application, exclude when raised

- In case of SLSTR-related reflectance bands:
  - cloud_an (see Table 8):
    - Exclude when not zero

As the OLCI instrument and the SLSTR nadir viewing instrument have different swaths (1270 km for OLCI, 1400 km for SLSTR nadir), the footprint of the OLCI and SLSTR bands in the surface reflectances product is not the same. For pixels where there are OLCI observations but no SLSTR observations, the IdePix annotation flag layer (based on OLCI) indicates IDEPIX_INVALID. When using OLCI bands, these pixels should be discarded from further processing.

The IdePix pixel classification can provide useful information, e.g. on cloud or snow occurrences, when SLSTR bands are used, but there are two drawbacks:

(1) the OLCI and SLSTR acquisitions are not exactly simultaneous and have different viewing angles,
(2) the OLCI and SLSTR footprints are not the same.

The atmospheric correction is run over all pixels, regardless of the pixel classification. Atmospheric correction over clouds is unreliable and can yield out of range values. Surface reflectance values outside the physical range (Table 6) should be in any case discarded.
Furthermore, the performance of the atmospheric correction methodology leads to the following recommendation for both OLCI and SLSTR reflectance bands:

- AC_process_flag (see Table 11)
  - Exclude when AOT > 1 (Bit 2 set)
  - Exclude when SZA > 65° (Bit 3 set)

Each user can decide to take this recommendation into account, or not, depending on his/her own applications.

5.2 ACCESSING THE ANNOTATION FLAGS WITH PYTHON

5.2.1 Accessing the flag 'cloud_an'

The following code snippet illustrates how to access the annotation flag 'cloud_an' from a surface reflectance product.

```
import numpy as np
from netCDF4 import Dataset

inFile = 'cgl_TOC_201907010000_X30Y12_S3A_v2.0.0.nc'
pfile = Dataset(inFile)
cloud_an = pfile['cloud_an'][:]
print(type(cloud_an))
print(cloud_an.dtype)
print(cloud_an)
```

If the flag band contains missing values, a numpy masked array is returned with 'missing_value' being masked. If there is no pixel with the 'missing_value' within the flag band, a numpy ndarray is returned. If a masked array is not suitable, a ndarray can be forced by adding one line of code before accessing the NetCDF variable.

```
pfile.set_auto_mask(False)
cloud_an = pfile['cloud_an'][:]
```

Note that using the set_auto_mask method will affect all variables being read / written afterwards. Variables accessed prior to using the method are not affected.

5.2.2 Accessing the flag 'quality_flags'

The following code snippet illustrates how to access the annotation flag 'quality_flags' from a surface reflectance product. This layer is a 32-bit integer flag which should be interpreted as an unsigned 32-bit integer. Due to the constraints in the CF-1.6 standard, unsigned variables are not allowed and a signed equivalent is used while adding an additional "_Unsigned = true" attribute to the flag band.

```
inFile = 'cgl_TOC_201907010000_X30Y12_S3A_v2.0.0.nc'
pfile = Dataset(inFile)
quality_fg = pfile['quality_flags'][:].astype(np.uint32)
print(type(quality_fg))
print(quality_fg.dtype)
print(quality_fg)
```

This will create a numpy masked array with uint32 values, and missing_value being masked.

The other annotation layers, cloud_an, pixel_classif_flags and AC_process_flag, are encoded as integers, containing negative numbers. These layers can by processed as normal.

5.3 ACCESSING THE SLSTR VAA AND SAA BANDS – VAA_SLSTR AND SAA_SLSTR

Due to the wrong specifications of the VAA_slstr and SAA_slstr encoding in the metadata of the netCDF files (see sections 3.3.3 for more details), viewers as QGIS will show 'nodata' when the angle is above 180°. Consequently, some additional steps are needed to read in the correct values. The following code snippet illustrates how to bypass the difficulty:

```
nc_file = 'cgl_TOC_20250331013452_X31Y04_S3B_v2.3.4.nc'
with netCDF4.Dataset(nc_file) as ds1:
 ds1.set_auto_mask(False)     # Disable automasking
 ds1.set_auto_scale(False)    # Disable autoscaling
 # Load the data as uint16
 VAA_SLSTR_corrected = ds1.variables['VAA_slstr'][:].astype("uint16")
 # Apply scaling to get physical angels between 0 and 360°
 VAA_SLSTR_scale_factor= ds1.variables['VAA_slstr'].scale_factor
 VAA_SLSTR_corrected_scaled = VAA_SLSTR_corrected * VAA_SLSTR_scale_factor
 # with mask applied
 VAA_mask = np.ma.masked_equal(VAA_SLSTR_corrected, 32767).mask
 VAA_SLSTR_masked = np.ma.masked_array(VAA_SLSTR_corrected_scaled, VAA_mask)
```

A similar approach can be used for SAA_slstr.

5.4 SUITABILITY FOR SPECIFIC SECTORS AND APPLICATIONS

As the incident energy reflected by the continental surface after removal of atmospheric interferences, the surface reflectance is the input for developing higher-level surface geophysical parameters. For instance, in the Copernicus Land Monitoring Service, the global surface reflectances are the main source of information for assessing a number of vegetation properties (https://land.copernicus.eu/en/products/vegetation). Indeed, the LAI, the FAPAR, the FCOVER, the land surface phenology, but also the NDVI, are derived from the Sentinel-3 surface reflectances. In the Copernicus Climate Change Service, they are also used to calculate the surface albedo (https://cds.climate.copernicus.eu/datasets/satellite-albedo).

The surface reflectances can also help in the detection and characterization of Earth surface changes. For example, in the Copernicus Land Monitoring Service, the burnt areas detection algorithm ingests both OLCI and SLSTR surface reflectances. They could also be used for pixel classification, land cover and land cover changes mapping.

Consequently, the surface reflectances can serve a broad range of users in Europe and across the World in the field of environmental terrestrial applications.

6 REFERENCES

Clerc, S., C. Donlon, F. Borde, N. Lamquin, S.E. Hunt, D. Smith, M. McMillan, J. Mittaz, E. Woolliams, M. Hammond, C. Banks, T. Moreau, B. Picard, M. Raynal, P. Rieu and A. Guérou, Benefits and lessons learned from the Sentinel-3 Tandem Phase, *Remote Sensing*, 12(17), 2668, 2020, https://doi.org/10.3390/rs12172668

Rahman H. and G. Dedieu, SMAC: a simplified method for the atmospheric correction of satellite measurements in the solar spectrum. *Intern. J. Remote Sens.*, 15, 123-143, 1994.

Vermote, E.F., D. Tanré, J.L. Deuzé, M. Herman and J.-J. Morcrette, Second Simulation of the Satellite Signal in the Solar Spectrum, 6S: An Overview, IEEE Transactions on Geoscience and Remote Sensing, Vol. 35, No. 3, p. 675-686, 1997.

7 ANNEX 1: REQUIREMENTS

According to the applicable document [AD1], the requirements relevant for the surface Reflectance products are described below.

7.1 SPECIFIC TECHNICAL DETAILS AND REQUIREMENTS

PRODUCT SPECIFICATION	
Geometric Properties:	
Spatial resolution	300m
Geolocation precision	Better than 0.5 pixels
Coordinate position	Centre of the pixel
Geodetic datum	WGS84
Geographic projection	Regular latitude/longitude grid
Geographic coverage	Global

TOP-OF-CANOPY REFLECTANCES	
Top-Of-Canopy (TOC) spectral reflectance of the Sentinel-3 OLCI and SLSTR sensor bands	
Uncertainty (2-sigma)	Threshold: 5% Goal: 3%
Stability (per decade, i.e. 10 years)	Threshold: 3% Goal: 1.5%

7.2 FURTHER REQUIREMENTS

7.2.1 Output product composition

Products may contain various information layers and ancillary information – the base reference for product packages are the operational products as on 01.03.2023.

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7.2.2 Data Structure

Data coding[4] shall be compatible with the Global Land products as on 01.03.2023 and/or follow the INSPIRE specifications, where applicable.

Ancillary information shall be as currently used and include at least the following:

- The number of measurements per pixel used to generate any synthesis product
- The per-pixel date of the individual measurement or the start-end dates of the period covered
- Quality indicators, with explicit per-pixel identification of the cause of anomalous parameter result.

The product naming and filename conventions that are used in the Copernicus Global Land component production as on 01.03.2023 shall be followed. This may be adapted for complete product collections upon agreement with the contracting authority during the Framework.

7.2.3 Data format

To ensure interoperability with the current Global Land component (operational product data formats and archive data formats) and other Copernicus services, all datasets will be available in NetCDF. Additional format such as Cloud Optimized Geotiff (COG) or ZAR format can be proposed for production or could be requested by the Contracting Authority during the Framework Contract.

7.2.4 Uncertainties and Validation

Uncertainties indicated in the product specifications above follow the threshold and goal proposed by GCOS#245.

Uncertainties estimates should account for the error propagation uncertainty coming from input data though the retrieval algorithms as during the contract period, ESA plans to imbed uncertainties in the Sentinel-3 ground segment products and, for Sentinel-2, there is an offline tool to determine Level 1 uncertainties; these can be used for propagation in the production chain.

Validation of the products shall conform to at least the CEOS LPV standards. Wherever appropriate the bio-geophysical variables shall be validated and compared to CEOS CAL/VAL data sets and/or Ground-Based Observations for Validation (GBOV) of Copernicus Global Land component when biophysical parameters are available.

[4] Data coding is the provision for the number of bits, range of values, usage of reserved values, content of status map, conversion to physical values formulas.

7.2.5 Input data

Copernicus Sentinel data are available from https://dataspace.copernicus.eu

Bio-geophysical variable products should be based on common base reflectance data:

- Sentinel 3: Derived Top-of-Canopy reflectance may be brokered or produced as it is the case on 01.03.2023 under the CGLOPS contracts;
- Sentinel 2: the Global Land Sentinel-2 Global Mosaic (S2GM)[5] component provides temporal mosaic of surface spectral bands that can be brokered and/or directly used.

Up to and including 2019, products archives of the Global Land component have been based on SPOT VGT, PROBA-V, ENVISAT, MODIS, TOPEX/Poseidon, Jason-1, Jason-2, Jason-3, datasets, which are available through the https://dataspace.copernicus.eu.

Ancillary satellite data that is purchased through Copernicus and put at disposal of the Services is available through the Data Warehouse and will become available on the Copernicus Dataspace Ecosystem.

Ancillary data sets, other than satellite imagery described above, that might be required shall be the responsibility of the contractor.

7.2.6 Product delivery

Products shall be delivered to the Copernicus Land Component dissemination.

The Copernicus Data Space Ecosystem infrastructure is used to provide access to the final map products.

[5] https://land.copernicus.eu/en/products/global-image-mosaic

8 ANNEX 2: EXAMPLE METADATA DUMP

For input OLCI L1:

gdalinfo
S3A_OL_1_EFR____20210401T201354_20210401T201654_20240603T034348_0179_070_142_3420_MAR_R_NT_004.SEN3/Oa01_radiance.nc

Driver: netCDF/Network Common Data Format

Files :
S3A_OL_1_EFR____20210401T201354_20210401T201654_20240603T034348_0179_070_142_3420_MAR_R_NT_004.SEN3/Oa01_radiance.nc

Size is 4865, 4091

Metadata:

NC_GLOBAL#absolute_orbit_number=26680

NC_GLOBAL#ac_subsampling_factor=64

NC_GLOBAL#al_subsampling_factor=1

NC_GLOBAL#comment=

NC_GLOBAL#contact=ops@eumetsat.int

NC_GLOBAL#creation_time=2024-06-03T03:43:48Z

NC_GLOBAL#history= 2024-06-03T03:43:48Z: PUGCoreProcessor /repvol/JOBTMPDIR_s3a-s3pug-efr_165.8453_16906_9w10s2ag/WD_S3A_OL1-1617306136.691808_1617308783.234492_20240603032241/JobOrder.xml

NC_GLOBAL#institution=MAR

NC_GLOBAL#netCDF_version=4.2 of Apr 12 2021 08:07:42 $

NC_GLOBAL#processing_baseline=**OL__L1_.004.00.00**

NC_GLOBAL#product_name=S3A_OL_1_EFR____20210401T201354_20210401T201654_20240603T034348_0179_070_142_3420_MAR_R_NT_004.SEN3

NC_GLOBAL#references=S3IPF PDS 004.1 – i2r6 – Product Data Format Specification – OLCI Level 1, S3IPF PDS 002 – i1r8 – Product Data Format Specification – Product Structures, S3IPF DPM 002 – i2r9 – Detailed Processing Model – OLCI Level 1

NC_GLOBAL#resolution=[270 294]

NC_GLOBAL#source=IPF-OL-1-EO 06.19

NC_GLOBAL#start_time=2021-04-01T20:13:53.695523Z

NC_GLOBAL#stop_time=2021-04-01T20:16:53.660650Z

NC_GLOBAL#title=OLCI Level 1b Product, Radiance Oa01 Data Set

Oa01_radiance#add_offset=0

Oa01_radiance#ancillary_variables=Oa01_radiance_unc

Oa01_radiance#coordinates=time_stamp altitude latitude longitude

Oa01_radiance#long_name=TOA radiance for OLCI acquisition band Oa01

Oa01_radiance#scale_factor=0.01394646

Oa01_radiance#standard_name=toa_upwelling_spectral_radiance

Oa01_radiance#units=mW.m-2.sr-1.nm-1

Oa01_radiance#valid_max=65534

Oa01_radiance#valid_min=0

Oa01_radiance#_FillValue=65535

Corner Coordinates:

Upper Left (0.0, 0.0)

Lower Left (0.0, 4091.0)

Upper Right (4865.0, 0.0)

Lower Right (4865.0, 4091.0)

Center (2432.5, 2045.5)

Band 1 Block=1217x1023 Type=Uint16, ColorInterp=Undefined

NoData Value=65535

Unit Type: mW.m-2.sr-1.nm-1

Offset: 0, Scale:0.0139464596286416

For input SLSTR L1 file, example of **2018** file:

gdalinfo
S3B_SL_1_RBT____20180515T000510_20180515T000810_20200902T185511_0179_003_130_1800_LR1_R_NT_004.SEN3/S1_radiance_an.nc

Driver: netCDF/Network Common Data Format

Files:
S3B_SL_1_RBT____20180515T000510_20180515T000810_20200902T185511_0179_003_130_1800_LR1_R_NT_004.SEN3/S1_radiance_an.nc

Size is 512, 512

Metadata:

NC_GLOBAL#absolute_orbit_number=274

NC_GLOBAL#comment=

NC_GLOBAL#contact=s3ome@acri-st.fr

NC_GLOBAL#creation_time=20200902T185511Z

NC_GLOBAL#history= 2020-09-02T18:55:11Z: PUGCoreProcessor joborder.13047572.xml

NC_GLOBAL#institution=LR1

NC_GLOBAL#netCDF_version=4.2 of Jul 5 2012 17:07:43 $

NC_GLOBAL#product_name=S3B_SL_1_RBT____20180515T000510_20180515T000810_20200902T185511_0179_003_130_1800_LR1_R_NT_004.SEN3

NC_GLOBAL#references=S3IPF PDS 005.1 – i2r9 – Product Data Format Specification – SLSTR Level 1, S3IPF PDS 002 – i1r7 – Product Data Format Specification – Product Structures, S3IPF DPM 003 – i1r8 – Detailed Processing Model – SLSTR Level 1

NC_GLOBAL#resolution=[500 500]

NC_GLOBAL#source=**IPF-SL-1 06.17**

NC_GLOBAL#start_offset=24014

NC_GLOBAL#start_time=2018-05-15T00:05:09.624701Z

NC_GLOBAL#stop_time=2018-05-15T00:08:09.317466Z

NC_GLOBAL#title=S3 SLSTR L1 Radiance and Brightness Temperatures Product. (Measurements + Annotations)
NC_GLOBAL#track_offset=1996

For input SLSTR L1: example of **2021** file:

gdalinfo
S3B_SL_1_RBT____20210403T152605_20210403T152905_20210405T003739_0179_051_025_3780_LN2_O_NT_004.SEN3/S1_radiance_an.nc/S1_radiance_an.nc

Driver: netCDF/Network Common Data Format

Files:
S3B_SL_1_RBT____20210403T152605_20210403T152905_20210405T003739_0179_051_025_3780_LN2_O_NT_004.SEN3/S1_radiance_an.nc

Size is 512, 512

Metadata:

NC_GLOBAL#absolute_orbit_number=15312

NC_GLOBAL#comment=

NC_GLOBAL#contact=eosupport@copernicus.esa.int

NC_GLOBAL#creation_time=20210405T003739Z

NC_GLOBAL#history= 2021-04-05T00:37:39Z: PUGCoreProcessor /data/ipf-s3/workdir14/614791655/JobOrder.614791655.xml

NC_GLOBAL#institution=LN2

NC_GLOBAL#netCDF_version=4.2 of Mar 13 2018 10:14:33 $

NC_GLOBAL#product_name=S3B_SL_1_RBT____20210403T152605_20210403T152905_20210405T003739_0179_051_025_3780_LN2_O_NT_004.SEN3

NC_GLOBAL#references=S3IPF PDS 005.1 – i2r9 – Product Data Format Specification – SLSTR Level 1, S3IPF PDS 002 – i1r7 – Product Data Format Specification – Product Structures, S3IPF DPM 003 – i1r8 – Detailed Processing Model – SLSTR Level 1

NC_GLOBAL#resolution=[500 500]

NC_GLOBAL#source=**IPF-SL-1 06.17**

NC_GLOBAL#start_offset=131205

NC_GLOBAL#start_time=2021-04-03T15:26:05.545927Z

NC_GLOBAL#stop_time=2021-04-03T15:29:05.239792Z

NC_GLOBAL#title=S3 SLSTR L1 Radiance and Brightness Temperatures Product. (Measurements + Annotations)

NC_GLOBAL#track_offset=1996

For input SLSTR L1: example of **2024** file. The processing baseline (SL__L1_.004.07.00) information is only present in the metadata since 28 February 2022.

gdalinfo
S3B_SL_1_RBT____20240923T235741_20240924T000041_20240924T012013_0180_098_030_4140_PS2_O_NR_004.SEN3/S1_radiance_an.nc

Driver: netCDF/Network Common Data Format

Files:
S3B_SL_1_RBT____20240923T235741_20240924T000041_20240924T012013_0180_098_030_4140_PS2_O_NR_004.SEN3/S1_radiance_an.nc

Size is 512, 512

Metadata:

NC_GLOBAL#absolute_orbit_number=33412

NC_GLOBAL#comment=

NC_GLOBAL#contact=eosupport@copernicus.esa.int

NC_GLOBAL#creation_time=20240924T012013Z

NC_GLOBAL#history= 2024-09-24T01:20:13Z: PUGCoreProcessor JobOrder.4621437.xml

NC_GLOBAL#institution=PS2

NC_GLOBAL#netCDF_version=4.2 of Jan 13 2023 10:05:23 $

NC_GLOBAL#processing_baseline=**SL__L1_.004.07.00**

NC_GLOBAL#product_name=S3B_SL_1_RBT____20240923T235741_20240924T000041_20240924T012013_0180_098_030_4140_PS2_O_NR_004.SEN3

NC_GLOBAL#references=S3IPF PDS 005.1 - i2r11 - Product Data Format Specification - SLSTR Level 1, S3IPF PDS 002 - i1r8 - Product Data Format Specification - Product Structures, S3IPF DPM 003 - i1r12 - Detailed Processing Model - SLSTR Level 1

NC_GLOBAL#resolution=[500 500]

NC_GLOBAL#source=**IPF-SL-1 06.22**

NC_GLOBAL#start_offset=55395

NC_GLOBAL#start_time=2024-09-23T23:57:41.392052Z

NC_GLOBAL#stop_time=2024-09-24T00:00:41.085940Z

NC_GLOBAL#title=S3 SLSTR L1 Radiance and Brightness Temperatures Product. (Measurements + Annotations)

NC_GLOBAL#track_offset=1996